

# Mujtaba Hussain Mirza

PhD Candidate in Computer Science | Sapienza University of Rome

📍 Rome, Italy | [Website](#) | [Google Scholar](#) | [LinkedIn](#) | [GitHub](#)

## Research Profile

Final-year PhD candidate in Computer Science at Sapienza University of Rome, supervised by [Prof. Iacopo Masi](#) at the [OmnAI Lab](#). My research seeks to advance the **security, safety, robustness, and trustworthiness** of AI systems. Within these broader areas, I work on adversarial robustness, machine unlearning, test-time adaptation, model hallucinations, and multimedia forensics.

## Education

### PhD in Computer Science

*Sapienza University of Rome*

**Supervisor:** Prof. Iacopo Masi

Research Area: Adversarial machine learning.

**Nov 2023 – Present**

*Rome, Italy*

### MSc in Computer Science

*Sapienza University of Rome*

Grade: 110/110 **cum laude**

**Sep 2021 – Oct 2023**

*Rome, Italy*

**Thesis:** *Probing the Energy Landscape of Discriminative Classifiers with Adversarial Perturbations.*

### BTech in Computer Science and Engineering

*University of Kashmir*

**CGPA:** 8.01/10.

**Sep 2015 – Dec 2019**

*Kashmir, India*

**Thesis:** *Stock Price Movement Prediction Using Market Indicators and Twitter Sentiment Analysis.*

## Publications

- **M. H. Mirza**, M. R. Briglia, S. Beadini, and I. Masi. “Shedding More Light on Robust Classifiers under the Lens of Energy-Based Models.” *European Conference on Computer Vision (ECCV)*, 2024.
- **M. H. Mirza**, A. D’Orazio, O. Melamed, and I. Masi. “A Provable Energy-Guided Test-Time Defense Boosting Adversarial Robustness of Large Vision-Language Models.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- M. R. Briglia, **M. H. Mirza**, G. Lisanti, and I. Masi. “Why Adversarially Train Diffusion Models?” *International Conference on Learning Representations (ICLR)*, 2026.
- G. Daidone, F. Bartolucci, M. R. Briglia, **M. H. Mirza**, G. Lisanti, and I. Masi. “Neutralizing Proactive Defense using Diffusion-based Upsampling.” *ACM Workshop on Information Hiding and Multimedia Security (IH&MMSec)*, 2026.
- L. Mautino, **M. H. Mirza**, and I. Masi. “Correlating Adversarial Robustness with the Energy of Untrained Classifiers.” *International Conference on Machine Learning and Cybernetics (ICMLC)*, 2026.

## Professional Experience

### Associate Machine Learning Engineer

*Jocata Financial Advisory & Technology*

**Dec 2020 – Dec 2021**

*Hyderabad, India*

- Developed machine learning solutions for anti-money-laundering applications, improving model accuracy from 88% to 94%; contributed to MLOps adoption through MLflow and graph-based analysis with Neo4j.

### Junior Research Analyst

*Great Learning*

**Apr 2020 – Dec 2020**

*Bangalore, India*

- Developed technical content on machine learning and Python for the Great Learning Blog and Great Learning Academy.

## Academic Service and Activities

### Workshop

Co-organizer, Workshop and Challenge on Unlearning and Model Editing (U&ME), ECCV 2026. [\[Link\]](#)

### Organization

### Peer Review

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), IEEE Transactions on Information Forensics and Security (TIFS), The Visual Computer (TVC).

### Student Supervision

Co-supervised a BSc thesis on artistic style erasure in Stable Diffusion using machine unlearning techniques.

### Research

### Presentations

Selected for oral presentation at IEEE-EURASIP Summer School on Signal Processing (S3P 2024), Capri, Italy; presented a poster at CISP–ELLIS Summer School 2025 and received full funding for participation.

### Conference Service

Volunteer, International Conference on Image Analysis and Processing (ICIAP), Italy.

## Technical Skills

### Programming & ML

Python, C/C++, PyTorch, Keras, scikit-learn, Hugging Face, OpenCV, NumPy, Pandas, and PySpark.

### Tools & Systems

MLflow, Git, Docker, Linux, AWS, SQL, PostgreSQL, Neo4j, and Jupyter.

## Awards & Funding

- Sapienza University PhD Scholarship, awarded by Sapienza University of Rome through competitive selection (2023).
- ELSA (European Lighthouse on Secure and Safe AI) Travel Grant for participation in the CISP–ELLIS Summer School (2025).

## Interests

Beyond research, I enjoy blogging (my first paid job), traveling (a habit I picked up after moving to Europe), cooking (where my experiments usually end in a delicious dinner), and hiking (perhaps a consequence of growing up in the mountains of Kashmir).